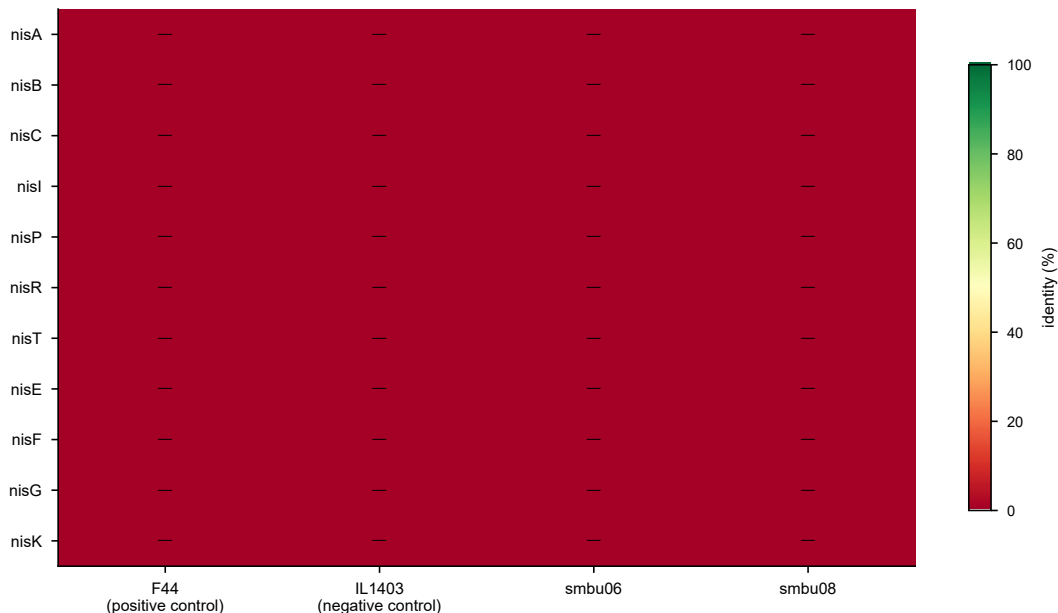


Nisin reference proteins (tblastn, best identity %)
cluster call: complete = nisA+nisB+nisC ≥80% identity / ≥80% qcov



F44 positive control: complete nisin cluster (nisA 92.98%, nisB/C/I/P/R/T 99.6–100%), IL1403 negative control: not detected.
smbu06/smbu08: not detected — only generic ABC-family homology (nisF 45.81%, nisT 34.45%) and a weak nisR-family response-regulator hit (34.96%);
the F44 nisin-cluster nucleotide region (14,401 bp) has zero BLASTN hits in either genome. Conclusion: no complete nisin locus in smbu06/smbu08.